



Rounak Gupta
Chemical Engineering
Indian Institute of Technology Bombay

22B2474
B.Tech.
Gender: Male
DOB: 25/12/2003

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2026	Core: 173 Total: 227
Intermediate	CBSE	Jusco School South Park	2022	94.60%
Matriculation	ICSE	Kerala Public School Gamharia	2020	96.20%

Pursuing a **Minor degree** in the Department of **Computer Science and Engineering**, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Achieved **City Rank 3** in 2014, School Rank 1 in **3 consecutive** years in SOF **International Mathematics Olympiad**
- Ranked **top 15 nationally** at IPTSE Olympiad in **Intellectual Property**; honored by **Chief Justice Prathiba Singh**
- Awarded **Student of the Year** by **L.I.C India** (2013) and **All-Rounder Trophy** in 12th grade for excellence (2022)
- Achieved **99.5 percentile** among **1.02M** JEE aspirants and **WBEE AIR 909**, demonstrating academic excellence

INTERNATIONAL EXPOSURE

International Academic Research Exchange Program | Institut Polytechnique de Paris [Jan'26-June'26]

- Charpak Scholar (Govt. of France)**; selected among **top 15** in India based on merit, **LoR**, research, projects, SOP
- Selected among **top 2** from IIT Bombay for **École Polytechnique, IP Paris(QS Top 40)** for research based on merit
- Research course in Applied Mathematics, **Tech, Finance & Blockchain Economics**; focus **FinTech** innovations

PROFESSIONAL EXPERIENCE

Data Science and Artificial Intelligence Intern | National Payments Corporation of India [May - June'25]

Received a **Pre-Placement Offer** for delivering high-impact AI solutions with strong ethics and team bonding

Data Analyst Intern | NoQs Digital

[Dec'23 - Jan'24]

- Analyzed **MSME data** to optimize operations & improve customer acquisition, increasing engagement by **18%**
- Extracted actionable insights from **complex datasets** using Power BI, Excel & Tableau, boosting decisions by **25%**
- Developed dashboards & **automated workflows** using **Form Mule & Autocrat**, saving **1k+ hours** annually for firm

KEY TECHNICAL PROJECTS

Full Mern Stack Web Development | Seasons of Code | Web n Coding Club, IIT Bombay [May'24-July'24]

- Developed full-stack MERN application using React, Node.js, Express, MongoDB, improving **UI load speed by 30%**
- Engineered secure backend with JWT auth, optimized MongoDB for **500+ CRUD ops/sec** and integrated **Redux**
- Deployed production-ready app on Heroku with CI/CD pipelines, supporting **1k+** concurrent users & smooth API calls

Deep Reinforcement Learning for Stock Market Trading | FinSearch | Finance Club, IIT Bombay [July'24]

- Developed **DQN, DDPG** trading agents for **NIFTY50**, boosting annualized returns **18.4%**, cutting drawdown **12%**
- Processed **10 years** of market data with custom state/action/reward design, thereby raising **Sharpe ratio by 0.27**
- Built **end-to-end RL pipeline** in Python/TensorFlow with backtesting & live testing, for consistent risk-adjusted gains

Build Your Own GPT Bot | Seasons of Code | Web n Coding Club, IIT Bombay

[May'25 - July'25]

- Developed **GPT-style chatbot** pipeline from tokenization → **BERT/LSTM** embeddings, boosting sentiment by **4%**
- Engineered **RAG workflows** with **LangChain and FAISS**, enabling context-aware **IITB law & academic bots >90%**
- Deployed **local, cloud bots** via **Gemini API and Ollama**, benchmarking **latency, hallucinations, and privacy**

Out-of-Distribution-Detection | CS419 Course Project | Prof. Abir De (Computer Science) [Spring'24]

- Implemented OOD detection on **CIFAR-10** using CNNs, boosting anomaly classification accuracy to **90% reliably**
- Enhanced detection using **Softmax Thresholding & OpenMax**, leveraging deep learning methods for high reliability
- Conducted **exhaustive parameter tuning** and experiments, achieving **90% OOD detection** with robust metrics

- Solved **120+** algorithmic problems with **85% acceptance**, **ranking top 10%** through optimized C++ solutions
- Mastered STL, Graphs, DP, Greedy & Network Flows to deliver efficient solutions across diverse problem sets

- Built and backtested **ARIMA** & cointegration strategies, generating **12.4% annualized** return & **63 percent** win rate
- Applied **derivatives pricing** and **hedging models**, reducing portfolio **risk by 8%** via futures, forwards, MTM strategy

- Solved **100+** DSA problems in C++ on arrays, trees, graphs & DP, boosting **problem-solving speed** significantly
- Applied **time & space optimization** across solutions, enhancing coding interview accuracy and readiness by **2×**

- Achieved **94% F1 POS** & **98.9% NER** by building **HMM**, **CRF** taggers and **SVM** with custom feature engineering
- Developed interactive Gradio/Streamlit UIs for live POS/NER tagging, enabling real-time evaluation and demos
- Fine-tuned **Phi-3.5 via DPO** on **2k** samples, boosting **Rouge-L vs baseline** under **limited GPU** compute efficiently

- Achieved **96 percent precision** detecting fraud on **284K+ credit card** transactions using Isolation Forest model
- Implemented **K-Means clustering** for segmentation and linear regression (**R² = 0.86**) and optimized key features
- Analyzed **complex video game dataset** with NumPy, Pandas, Matplotlib, Seaborn, extracting **actionable insights**

- Implemented **K-Means + PCA** on **9,458 NHANES records (27 variables)**, achieving **silhouette score 0.69**
- Engineered **risk-scoring framework** integrating disease, meds & lifestyle, enhancing **3-tier ML** model interpretability

- Developed **Arduino & ESP32** system to capture and transmit **pH sensor** data, boosting accuracy by **20%**
- Enabled **remote monitoring** with dashboards for reaction tracking, improving efficiency of data use by **30%**

- Developed **simulation pipelines** for feature extraction, captioning, and fluid modeling, improving **efficiency 20%**
- Applied **Beam Search** decoding on **LSTM** outputs, boosting **BLEU scores by 15%**, enhancing caption generation

POSITIONS OF RESPONSIBILITY

- Orchestrated **10+** large-scale initiatives, leading **500+ volunteers**, impacting **1M+** people nationwide and NGOs
- Directed **flagship** event Flare for **2K+** attendees, negotiated **10+ artist** deals saving **Rs.0.3M**, drove eco-donation

TECHNICAL SKILLS

- **Programming & Development**

: Python, C++, Java, JSON, HTML, $\LaTeX$, JavaScript, Excel, CSS, Flutter, LLM
- **Packages/Libraries**

: PyTorch, Pandas, TensorRT, Keras, Numpy, OpenCV, PyGame, SciPy, Sci-kit

RELEVANT COURSES UNDERTAKEN

Technical	Makerspace, Computer Programming, Logic for Computer Science, AI & Data Science, Machine Learning, Speech & Natural Language Processing, Computational Methods Lab, Computer Networks, Public Health Informatics, Numerical Analysis, Finance, Blockchain economics
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EXTRA-CURRICULAR ACTIVITIES

Competition & Sports	• Part of Inter-IIT Cricket-Summer Camp (2023) Played volleyball as hobby for 6 years • Part of Inter-IIT Cultural's'23, performed theatre infront of 6K+ audience, led coordination • Selected in top 5 in IITB to pitch at EMC Global Competition, competing with Japan scholars • Awarded top 9 Best Design Project by Worley at IChE competition in 2025, working in 6
Leadership & Pitching	• Mentored teams in Financial Mathematics & Cryptography and BMC competitions on AI • President, School Rotary Club, raised Rs.0.2M+ funds, educated orphanage children • Certified in Blockchain Shala Gen AI course in SOS Selected for ICETBM conference